Semantic Validation Gates: A Computable, Statistically Calibrated Framework for Runtime Verification of Language-Model Outputs

Luoqing Zhao
Independent Researcher

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Abstract

As language models and autonomous agents produce long reasoning chains and consequential decisions, deployment requires a verification layer that converts qualitative judgments—well-formed, factual, consistent, safe, on-task—into calibrated, auditable measurements. We present a five-gate runtime validation framework in which every object is computable from real model outputs and every guarantee is finite-sample with stated assumptions. The gates are: an exact edit-distance format gate; a calibrated-entailment fact gate with mandatory abstention; a logic gate built on a *weighted cellular sheaf Laplacian* over the extracted claim graph, unified with propositional constraints through a relax-and-certify architecture whose necessity we prove via a complexity-theoretic incompleteness result; a safety gate that is computationally independent of all parsing; and an intent gate whose aggregation form is forced by a dilution theorem. Thresholds are calibrated by conformal methods under three deployment regimes—exchangeable, covariate-shifted, and arbitrarily (adversarially) drifting—yielding end-to-end, instrument-inclusive false-rejection control. The decision layer evaluates all gates in parallel and grants alignment violations an unconditional veto, provably eliminating the masking of safety failures by lower-priority errors. A mechanism study on synthetic reasoning chains shows that as the distance k between a violation and the claim it contradicts grows, threshold-based contradiction counting collapses to chance (AUROC 0.50 at k=5) while the sheaf energy degrades gracefully (AUROC 0.97) and retains 83% top-1 localization where counting retains 0%. We position the framework as the missing theory layer for guardrail pipelines already deployed in production without statistical guarantees, and release a reference implementation and a pre-registered protocol for full empirical validation.

1 Introduction

1.1 Problem

Production systems increasingly interpose *guardrail pipelines* between a language model and its consumers: stacks of validators for output structure, factuality, consistency, safety, and task adherence. These pipelines are effective engineering but carry no mathematical contract: which validator fires first is a configuration accident, error taxonomies are informal, and false-rejection rates are tuned by feel. Conversely, the academic literature supplies deep tools for *individual* dimensions—conformal factuality, consistency detection, safety classification, instruction-following evaluation—that do not talk to each other. This paper supplies the missing layer: a single formal system in which five validation dimensions are computable scores, thresholds carry finite-sample guarantees under explicitly graded distributional assumptions, the induced error taxonomy is complete and mutually exclusive, and safety violations cannot be masked.

An earlier version of this work (v1) proposed the same five-gate architecture wrapped in differential-geometric language—Wasserstein projections onto convex sets, Čech obstruction classes with Hodge norms, isometric embeddings of Wasserstein geometry into Hilbert spaces—that was not constructible; Appendix A records each defect and its repair. The present version retains the architecture and delivers it with objects that exist.

1.2 Contributions

1. Five gate scores $f_1, \dots, f_5$ (Format, Fact, Logic, Alignment, Intent), each with an explicit algorithm, estimator, and sensitivity analysis (§3).

2. A logic gate built on a weighted cellular sheaf Laplacian over the extracted claim graph, unified with propositional constraints via relax-and-certify, together with an impossibility result (Prop. 3.3) explaining why a certification layer is unavoidable (§3.3).

3. Calibration with finite-sample guarantees across three deployment regimes: split conformal under exchangeability, weighted conformal under estimable covariate shift, and adaptive conformal inference whose long-run guarantee holds under arbitrary, including adversarial, drift (§4). The guarantees are *instrument-inclusive*: claim-extraction and NLI errors cannot inflate the false-rejection rate (Prop. 4.1).

4. A decision architecture with unconditional parallel evaluation and a safety veto, provably immune to masking of alignment violations (§5), together with an anytime computational profile whose safety fast path is a single classifier forward pass (§5.4).

5. A mechanism study demonstrating, under two documented assumptions about NLI behavior, a structural separation between the sheaf energy and the standard contradiction-counting baseline on transitive violations, including a localization capability the baseline lacks by construction (§6); and a pre-registered protocol for full empirical validation (§7).

1.3 Design principles

(P1) Every object is mathematically well-defined and computable from real model outputs. **(P2)** Every guarantee is finite-sample, with its assumptions stated next to it. **(P3)** Signals of localized or safety-critical violations are aggregated by *max/veto*, never by mean or minimal-index truncation. **(P4)** Anything not meeting P1–P2 is labeled a design choice or an open problem.

1.4 Related work

Each pillar has independent precedents; the contribution here is the certified composition. *Sheaves for consistency:* Huntsman, Robinson and Huntsman [14] demonstrated that LLMs produce usable numerical consistency ratings of claim pairs and proposed sheaf theory for lifting local ratings to global consistency of hypertexts, also noting the relevance of CNF-SAT; our §3.3 differs in operating on a single output's claim graph, weighting the Laplacian by calibrated NLI confidences with a proven sensitivity bound, and embedding the score in a calibrated gate rather than a stand-alone analysis. *Conformal guarantees for LLMs:* conformal factuality [16], conformal abstention for hallucination control [20], boosted and conditional variants [3], and conformal factuality under covariate shift via online density-ratio estimation [4] calibrate a *single* dimension; §4 calibrates a five-dimensional gate vector jointly with family-wise control and a veto-aware fixed-sequence procedure. *Guardrail pipelines:* production frameworks (Guardrails AI, NeMo Guardrails [17], and similar) stack validators covering roughly our five dimensions but provide no statistical guarantees, no unified severity measure, and no masking analysis; this paper can be read as the theory layer for that architecture. *Multi-dimensional evaluation suites* such as HELM [13] systematize offline benchmark dimensions for scoring *models*; our object is different—runtime, per-output verification with per-instance guarantees and block/repair decisions.

2 Output Representation

Fix a deterministic encoder $E : \Sigma^* \to \mathbb{R}^d$ applied at unit level: an output y is segmented by a fixed procedure D into units (sentences or atomic claims) $u_1, \dots, u_m$.

Definition 2.1 (Output measure). $\hat{\mu}_y = \frac{1}{m} \sum_{j=1}^{m} \delta_{E(u_j)} \in \mathcal{P}(\mathbb{R}^d)$.

Assumption 1 (A1). *$\|E(u)\| \le R$ for all units (normalized embeddings: $R = 1$).*

Assumption 2 (A2). *All kernels are bounded ($\sup_x k(x,x) \le K$), measurable, and characteristic on the relevant domain.*

Assumption 3 (A3). *Calibration and test outputs are exchangeable* as raw texts. *Used only where stated; §4.4 removes it.*

A *gate* is a pair (f_i, τ_i) with measurable $f_i : \Sigma^* \to [0, \infty)$ and threshold $\tau_i > 0$; write $\rho_i(y) = f_i(y)/\tau_i$, and gate i passes iff $\rho_i(y) \le 1$. The *instrument* $I = (D, \mathrm{NLI}, E, \text{graph builder})$ is fixed before calibration and treated as part of each f_i; this matters for Prop. 4.1. No manifold, bundle, or Fisher-metric structure is assumed; everything downstream lives on $(\mathbb{R}^d, \|\cdot\|)$ and on finite graphs.

3 The Five Gates

3.1 Format gate f_1

Let the structural specification be a formal language $L(G)$ (context-free grammar, JSON-Schema compilation, or regular constraint set). Define

$$f_1(y) = \frac{d_{\text{edit}}(y, L(G))}{|y|}, \qquad d_{\text{edit}}(y, L) = \min_{z \in L} d_{\text{Lev}}(y, z).$$

For CFGs this is computable *exactly* by error-correcting Earley/CYK parsing [1] in $O(|G|^2|y|^3)$; for regular L, by product-automaton shortest path in $O(|A||y|)$. The gate is deterministic with zero estimation error, and $f_1(y) = 0 \iff y \in L(G)$. No Wasserstein projection is needed; the v1 operator $\mathrm{Proj}_{\mathcal{P}_{\text{form}}}$ is replaced by an exact combinatorial minimization with a known algorithm.

3.2 Fact gate f_2

Decompose y into claims $c_1, \dots, c_m$; retrieve evidence E_j from a verified knowledge base $\mathcal{K}$; let s_j be the *calibrated* entailment probability from an NLI model (isotonic or Platt calibration on held-out data; ECE reported). Per P3,

$$f_2(y) = 1 - \min_{1 \le j \le m} s_j,$$

with a mandatory *abstention state*: when retrieval confidence for a claim falls below a floor, the output is routed to abstention rather than scored—an unverifiable claim must not receive a numerical pass. (This removes the v1 ambiguity in which kernel-smoothing of a reference measure silently interpolated over missing knowledge.) Optionally, against a reference sample from $\mathcal{K}$ one may use the Sinkhorn divergence $S_\varepsilon(\hat{\mu}_y, \hat{\nu}_{\text{fact}})$, which metrizes weak convergence with dimension-robust sample complexity $O(n^{-1/2})$ [7]; plug-in W_2 degrades as $O(n^{-1/d})$ [6] and is therefore not used anywhere in this paper.

3.3 Logic gate f_3: weighted sheaf Laplacian, unified energy, relax-and-certify

Claim graph. $G = (V, E)$: vertices are claims; an edge (u, v) with relation type (entail / equiv / contradict) is added when the NLI model flags it, carrying its calibrated confidence $w_e \in [0, 1]$.

Weighted sheaf. Attach to G a cellular sheaf $\mathcal{F}$ [12]: stalks $\mathcal{F}(v) = \mathbb{R}^k$ (truth coordinates of claim v); restriction maps encode the relation (identity for entailment and equivalence, the negation lift $t \mapsto 1 - t$ for contradiction). With coboundary $\delta : C^0 \to C^1$ and $W = \operatorname{diag}(w_e)$, the *weighted sheaf Laplacian* is $L_{\mathcal{F}}(w) = \delta^\top W \delta$, and the raw score of an assignment x is $\langle x, L_{\mathcal{F}}(w)x\rangle^{1/2}$: a finite-dimensional Hodge-type quantity, computable in $O(|V|^3)$ worst case and by sparse solvers in practice. Global sections $\ker \delta = H^0(G; \mathcal{F})$ are exactly the edge-consistent assignments, so the score measures the obstruction to gluing the asserted claims into a global section—the v1 intuition, now on a specified finite object.

Proposition 3.1 (Lipschitz dependence on NLI confidences)**.** *If $\|x_v\| \le R$ and restriction maps have operator norm at most 1, then for confidence vectors w, w',*

$$\left|\langle x, L_{\mathcal{F}}(w)x\rangle - \langle x, L_{\mathcal{F}}(w')x\rangle\right| \le 4R^2\|w - w'\|_1.$$

Proof. The energy is $\sum_e w_e \|(\delta x)_e\|^2$ with $\|(\delta x)_e\| \le 2R$; subtract termwise. □

An out-of-distribution NLI perturbation therefore produces a bounded, quantified score perturbation, which composes with margin stability (Prop. 5.7); "garbage in, garbage out" is replaced by a sensitivity bound. The bound is verified numerically in the reference implementation.

Unified logic energy. Propositional constraints enter as sheaf data: lift Boolean assignments to $[0, 1]$ stalks on the factor graph of extracted constraints, with restriction maps encoding literal polarity; let X be the linear-relaxation polytope of the clauses. Define

$$f_3(y) = \min_{\substack{x \in X \\ x \equiv 1 \text{ on asserted claims}}} \langle x, L_{\mathcal{F}}(w)\, x\rangle^{1/2}.$$

The objective is a convex piecewise quadratic; projected gradient or interior-point methods find the global minimum.

Proposition 3.2 (Relaxation gap)**.** *(i) If the asserted constraint system is Boolean-satisfiable then $f_3(y) = 0$. (ii) The converse fails in general: LP/SDP integrality-gap instances (e.g. MAX-2-SAT [10]) yield fractional zeros for unsatisfiable systems.*

Proof sketch. (i) A Boolean satisfying assignment lies in X and has zero residual on every edge by construction of the restriction maps. (ii) Standard gap instances embed directly. □

Proposition 3.3 (No exact polynomial unification)**.** *A polynomial-time computable continuous score s with $s(y) = 0$ iff the asserted propositional system is Boolean-satisfiable would decide* SAT *in polynomial time. Hence, unless* P = NP*, every polynomial unified score is incomplete on the hard fragment.*

Relax-and-certify. Consequently a SAT/SMT *certifier* is retained—not as a parallel gate but as a certification layer: it verifies (rounds) near-zero relaxed optima and, on UNSAT, returns a deletion-minimal unsatisfiable core whose size is an exact, interpretable severity overriding a passing continuous score. The soft/hard dichotomy of earlier drafts dissolves into the standard relax-and-certify hierarchy of modern optimization. As a diagnostic, the edges carrying the largest weighted residual energy *localize* the inconsistency; localization is evaluated in §6. A lattice-valued unification via the Tarski Laplacian [8] is recorded as Open Problem 8.1.

3.4 Alignment gate f_4

$f_4(y) = \sigma^{-1}(\hat{p}_{\text{unsafe}}(y))$: the logit of a safety classifier calibrated on held-out labels (ECE reported), composed with the conformal layer of §4 so that a bounded false-rejection rate on safe outputs holds as a theorem. *Computational independence (structural requirement):* f_4 operates on the *raw text* y and does not factor through D, the claim graph, or format parsing; a format-corrupted output therefore cannot even computationally block its own safety evaluation. Combined with the veto of §5.2, the masking channel is closed at the dataflow level as well as the decision level. (The v1 formulation—W_2 projection onto a "compact convex set of safe behaviors"—is dropped: the set was never constructed and metric projection in positively curved Wasserstein space is not well-posed.)

3.5 Intent gate f_5

Proposition 3.4 (MMD dilution)**.** *Let $\hat{\mu}, \hat{\mu}'$ be empirical measures on m units differing in exactly one unit, with kernel bound $k \le K$. Then $\mathrm{MMD}_k(\hat{\mu}, \hat{\mu}') \le 2\sqrt{K}/m$.*

Proof. $\psi(\hat{\mu}) - \psi(\hat{\mu}') = \frac{1}{m}(\phi(e_j) - \phi(e'_j))$ and $\|\phi(\cdot)\| \le \sqrt{K}$. □

A fatal single-unit violation (one inserted negation) moves an aggregate-MMD score by $O(1/m)$—vanishing in long outputs. Hence, per P3,

$$f_5(y) = \max\bigl(f_5^{\text{hard}}(y),\, f_5^{\text{soft}}(y)\bigr).$$

Hard component (exact). Compile the instruction into programmatically checkable constraints $\{\kappa_\ell\}$ (length, inclusion/exclusion, ordering—the IFEval regime [15]); $f_5^{\text{hard}} = \max_\ell \mathbf{1}[\kappa_\ell \text{ violated}]$. A single negation that flips a checkable constraint scores 1 regardless of m. *Soft component (worst-unit, directed Hausdorff in feature space).*

$$f_5^{\text{soft}}(y) = \max_j \min_{r \in \mathrm{supp}(\nu_{\text{intent}})} \bigl\|\phi(E(u_j)) - \phi(E(r))\bigr\|_{\mathcal{H}_k},$$

where ν_{intent} is the reference measure of the task objective.

Proposition 3.5 (Unit-level sensitivity)**.** *If one unit lies at feature distance at least Δ from the entire intent reference set, then $f_5^{\text{soft}} \ge \Delta$, independently of m.*

The kernel mean embedding $\psi(\mu) = \int \phi\, d\mu$ is isometric *with respect to MMD* by construction, and MMD is a metric for characteristic k [11]. This replaces the v1 claim of an isometric RKHS embedding of the W_2 geometry, which is impossible for $d \ge 3$ [2]. Three honesty notes: (i) cross-encoder relevance scores are generally not positive semidefinite and may be used as raw scores, not kernels—conformal calibration is agnostic to the score's provenance; (ii) the $O(n^{-1/2})$ MMD concentration bound applies to the mean-embedding form; the max-form relies on conformal calibration for validity and on §7 for empirical power; (iii) aggregate MMD is retained solely as a *drift-monitoring* feature (§4.4), where distribution-level sensitivity is exactly what is wanted.

4 Calibration: Finite-Sample Guarantees Under Three Regimes

4.1 The guarantee is instrument-inclusive

Proposition 4.1 (Composite conformal validity)**.** *Fix the instrument I before calibration and let $f_i = g_i \circ I$. If calibration and test outputs are exchangeable as raw texts (A3), then Theorem 4.2 below holds verbatim for the composite score.*

Proof. Exchangeability is preserved under a fixed measurable map applied identically to every element of the sequence; the rank argument of Theorem 4.2 then applies to the composite scores unchanged. □

Claim-extraction truncation and out-of-distribution NLI behavior therefore *cannot* inflate the false-rejection rate: instrument noise enters calibration and test exchangeably. What instrument error does affect is detection *power* (Type-II), quantified in §4.5 and audited in §7.

4.2 Regime I — exchangeable deployment: split conformal

With a calibration set of n_i outputs certified valid on dimension i, set τ_i to the $\lceil (n_i+1)(1-\alpha_i) \rceil$-th smallest calibration score.

Theorem 4.2 (Distribution-free false-rejection control)**.** *Under (A3), for a new valid output y, $\mathbb{P}(\rho_i(y) > 1) \leq \alpha_i$; if scores are almost surely distinct, also $\mathbb{P}(\rho_i(y) > 1) \geq \alpha_i - \frac{1}{n_i+1}$.*

Proof. By exchangeability, the rank of $f_i(y)$ among the $n_i + 1$ scores is uniform on $\{1, \ldots, n_i + 1\}$ (ties only help the upper bound). Exceedance of τ_i requires the rank to exceed $\lceil (n_i+1)(1-\alpha_i) \rceil$, an event of probability at most α_i; the lower bound follows from the same rank computation without ties. □

This is precisely the property v1's "EVT conjecture" asked for, obtained as a three-line finite-sample theorem with no distributional assumptions. (Extreme-value refinements remain optional; note that all gate scores here are bounded, placing them in the *Weibull*, not Fréchet, max-domain of attraction [5]—any EVT use must carry domain diagnostics.)

Theorem 4.3 (Family-wise validity, safety-first)**.** *Choosing $\alpha_i = \alpha/5$ yields $\mathbb{P}(\exists i : \rho_i(y) > 1 \mid y \text{ fully valid}) \leq \alpha$ by the union bound. Where a fixed-sequence procedure is used to recover power under dependence, gates are tested* safety-first *(f_4 at full level α before all others), aligning power allocation with the veto semantics of §5.2.*

4.3 Regime II — estimable covariate shift: weighted conformal

When the shift is estimable (e.g. a changed prompt mix), weighted conformal prediction [18] restores the per-instance guarantee with likelihood-ratio weights $\hat{w}(x)$; weight-estimation error enters as an explicit total-variation correction. (Online density-ratio estimation as in [4] is a compatible single-gate instance.)

4.4 Regime III — arbitrary/adversarial drift: adaptive conformal inference

Deployment default: the online update of Gibbs and Candès [9], $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$, with the conformal quantile taken at level $1 - \alpha_t$.

Theorem 4.4 ([9], restated)**.** *For arbitrary—including adversarially chosen—sequences of distributions,* $\left|\frac{1}{T}\sum_{t\le T}\mathrm{err}_t - \alpha\right| \le \frac{\max(\alpha_1, 1-\alpha_1)+\gamma}{\gamma T} \to 0.$

Long-run false-rejection frequency converges to nominal α with *no* exchangeability or stationarity assumption; the guarantee does not lapse between recalibrations because adaptation is per-step—there is no "blind window" at the level of long-run frequency. (A3) is thereby demoted from a load-bearing assumption to the condition under which the stronger per-instance bound of Theorem 4.2 additionally holds.

Monitoring and fail-safe posture. Drift monitoring uses conformal test martingales / e-processes [19]: anytime-valid, no scheduled testing lag, false-alarm control at every stopping time; the monitored features include the aggregate-MMD statistic of §3.5. Upon alarm and until re-validation, thresholds are tightened monotonically and low-margin outputs are routed to abstention or human review.

Proposition 4.5 (Safe-side degradation)**.** *Decreasing τ_i enlarges every rejection region and shrinks none; during the alarm state the probability of a false* accept *on any dimension is non-increasing relative to the pre-alarm configuration.*

Proof. $\{f_i > \tau_i\}$ is decreasing in τ_i under set inclusion. □

What is sacrificed under drift is throughput (more abstentions), never the safety direction of the error.

4.5 Power accounting

Proposition 4.6 (Detection power lower bound)**.** *If an inserted violation survives claim decomposition with probability at least $1-\varepsilon_D$, is flagged as an edge by NLI with probability at least $1-\varepsilon_{\mathrm{NLI}}$ given survival, and, once in the graph, drives $\rho_3 > 1$ with probability at least π_3, then $\mathbb{P}(\textit{detected}) \ge (1-\varepsilon_D)(1-\varepsilon_{\mathrm{NLI}})\,\pi_3$.*

ε_D and $\varepsilon_{\mathrm{NLI}}$ are estimated on the audit benchmarks of §7 with Clopper–Pearson intervals and enter the reported power certificate. *Abstention rule:* if extraction coverage (fraction of tokens attributable to extracted claims) falls below a floor $c_{\min}$, the output is routed to abstention.

4.6 Reporting requirements

For every deployment: n_i; empirical coverage on held-out valid data with Clopper–Pearson intervals; ECE for every internal classifier (f_2's NLI, f_4's safety classifier); the e-process monitor state; and the regime (I/II/III) under which each guarantee is claimed.

5 Decision Architecture: Partition, Veto, Stability

5.1 Parallel evaluation and the partition theorem

All five scores are computed for every output, unconditionally; no gate's computation may be short-circuited by another's failure. The primary logged object is the *failure vector* $K(y) = \{i : \rho_i(y) > 1\}$.

Lemma 5.1 (Measurability). *Each f_i of §3 is measurable: f_1 is computable; f_2, f_4 are compositions of measurable maps with continuous links; f_3 is continuous in the assignment for each fixed graph, with graph extraction a measurable map into a countable set of structures; f_5 is a max/min of continuous functionals. Hence all maps below are measurable.*

Theorem 5.2 (Completeness and mutual exclusivity). *For any finite family of measurable gates, with $\Pi(y) = \min K(y)$ (and 0 if $K(y) = \emptyset$),*

$$\Sigma^* = V_{\text{valid}} \sqcup \bigsqcup_{i=1}^{5} E_i, \qquad E_i = \{y : \rho_i(y) > 1,\ \rho_j(y) \le 1\ \forall j < i\}.$$

Proof. Well-ordering of the failure set $K(y)$. □

Remark 5.3 (Honesty). Theorem 5.2 holds for *any* five measurable criteria and certifies nothing about gate quality; it is bookkeeping. The substantive guarantees are Theorems 4.2–4.4 and the propositions of this section.

5.2 Safety veto and the demotion of Π

$$\mathrm{Dec}(y) = \begin{cases} \textsf{Block} & 4 \in K(y) \quad \text{(safety veto, order-independent)} \\ \textsf{Repair via } \Pi_{-4}(y) & K(y) \neq \emptyset,\ 4 \notin K(y) \\ \textsf{Pass} & K(y) = \emptyset \end{cases}$$

where $\Pi_{-4}(y) = \min K(y)$ over the non-veto gates. Π survives *only* as a repair-routing heuristic ("fix the format before re-checking facts"), a role in which lexicographic order is defensible; it carries no reporting or risk semantics.

Proposition 5.4 (No masking). *Under* Dec, *every output with $\rho_4(y) > 1$ is Blocked regardless of $\rho_1, \rho_2, \rho_3, \rho_5$. The scenario "$f_1$ fails first and suppresses an f_4 violation" is unrealizable: "first" does not exist at the decision layer, and by §3.4 it does not exist at the dataflow layer either.*

Remark 5.5 (Π is not a security boundary). Consider an adversary who deliberately plants a mild fact violation so that f_2 "fails first." Against a sequential, short-circuiting evaluator this attack masks a co-occurring alignment violation—it is valid against the v1 semantics. Under the present architecture it accomplishes nothing: all five scores are computed unconditionally, $K(y)$ records both failures, f_4 reads the raw text, and Dec blocks on $4 \in K(y)$ irrespective of every other entry. Mutual exclusivity is a property of the *reporting taxonomy* Π (one primary label per output); diagnostic completeness is carried by $K(y)$, which is always full. The security boundary is Dec, never Π.

5.3 Well-posedness and stability

Proposition 5.6 (Almost-sure well-posedness). *If each $f_i(y)$ has an atomless law at τ_i under the deployment distribution, then almost surely every output has strictly positive margin $\gamma(y) = \min_i |f_i(y) - \tau_i|$, and Π, K,* Dec *are locally constant under score perturbations smaller than $\gamma(y)$.*

The condition is empirically checkable via continuous score histograms; this replaces the v1 conjecture that invoked a (non-existent) canonical reference measure on Wasserstein space and a Hausdorff-dimension bound that is meaningless in infinite dimensions.

High dimensions and measure concentration. The classification factors through the five scalar scores: Dec depends on y only via $(f_1(y), \ldots, f_5(y)) \in \mathbb{R}^5$. High-dimensional geometry of the embedding space therefore enters only through the one-dimensional law of each score; no claim about the Hausdorff dimension or measure of a boundary in $\mathcal{P}(\mathbb{R}^d)$ is made or needed—the v1 boundary conjecture is not repaired but *dissolved*. The legitimate residue of the concentration concern is empirical: score laws might concentrate *near* τ_i, producing many small-margin outputs. Two answers. First, the validity guarantees (Theorems 4.2 and 4.4) hold regardless of concentration; margins matter only for estimation stability (Prop. 5.7). Second, we mandate reporting the empirical margin distribution $\hat{\mathbb{P}}(|f_i - \tau_i| \leq \delta)$ and support an optional *abstention band*: outputs with margin below δ are routed to review, converting boundary ambiguity from a silent failure mode into a measured abstention rate.

Proposition 5.7 (Margin-based stability of estimated classification)**.** *If simultaneously* $|\hat{f}_i - f_i| \leq \varepsilon_i$ *for all* i *with probability at least* $1 - \delta$ *(MMD concentration for the mean form;* $\varepsilon_1 = 0$*; Prop. 3.1 converting NLI-confidence error into* ε_3*) and* $\gamma(y) > \max_i \varepsilon_i$*, then* $\widehat{\mathrm{Dec}}(y) = \mathrm{Dec}(y)$ *with probability at least* $1 - \delta$*.*

Remark 5.8 (Stability–sensitivity separation). Two perturbation classes must not be conflated. *Measurement perturbations* (sampling noise, NLI-confidence noise) should not flip decisions; Prop. 5.7 delivers this above the margin. *Semantic perturbations* (inserting a single negation) *should* flip decisions, and the gates are engineered to be non-smooth in exactly this direction: a single-unit flip at feature distance Δ moves f_5^{soft} by at least Δ (Prop. 3.5) and moves the min/max aggregations of f_2 and f_5^{hard} by $O(1)$, while measurement noise moves scores by at most ε; the two regimes are distinguishable whenever $\Delta > 2\varepsilon$. Local constancy is claimed in the *score metric*, never in the semantic metric; demanding the latter would reintroduce the dilution pathology that Prop. 3.4 rules out.

Proposition 5.9 (One-sided metric comparison)**.** *For* L*-Lipschitz bounded* k *and measures supported in the* R*-ball,* $\mathrm{MMD}_k \leq C(L, K)\, W_1 \leq C(L, K)\, W_2$*. The converse inequality fails in general; only this direction is used, and only for interpretability (small transport perturbations of an output cannot inflate the soft intent score), never for a guarantee.*

This replaces the v1 conjecture of global Lipschitz *equivalence* of W_2, harmonic, and RKHS norms, which is false as stated.

5.4 Computational profile and anytime operation

Latency is graded per gate, and the architecture guarantees that safety never waits for expensive computation. *Fast path (synchronous, per output):* f_4 is one classifier forward pass on the raw text, and f_1 is linear-time validation (membership checking), with the cubic error-correcting parse invoked only on failure to quantify severity; the veto of §5.2 therefore executes within a single forward-pass latency. *Slow path (deadline-bounded):* f_2 is dominated by retrieval and NLI passes; the relaxed energy of f_3 is a sparse convex program on a claim graph with m vertices (m in the tens for typical outputs), i.e. millisecond-scale linear algebra, and is updatable *incrementally* under streaming generation by low-rank updates of $L_{\mathcal{F}}(w)$ as claims and edges arrive. The SAT certifier is invoked lazily under a deadline: the relaxed score is an *anytime* answer, and if the deadline expires the output carries an "uncertified" flag with conservative handling. This is a principled rather than an apologetic design: Prop. 3.3 shows the worst-case certification cost is unavoidable for *any* exact method, so the only honest options are anytime relaxation plus optional certification—which is what the gate does. *Scope honesty:* microsecond decision loops (e.g. high-frequency trading) are outside the scope of any semantic verification—no entailment model evaluates in microseconds; the target regime is the one in which generation itself costs 0.1–10 s and full verification runs at a fraction of that budget, with the safety fast path always inside it.

6 Mechanism Study: Sheaf Energy versus Contradiction Counting

The framework's mathematical identity rests on one empirical question: does the sheaf energy f_3 add signal beyond the trivial baseline that counts NLI-flagged contradiction edges above a decision threshold? Before committing to full-scale experiments (§7), we isolate and demonstrate the *mechanism* by which a separation should arise, under two documented assumptions about NLI behavior:

(S1) *Distance decay.* The confidence assigned by a pairwise NLI model to the direct contradiction edge decays with the semantic/reasoning distance k between the violating statement and the claim it contradicts; in the simulation, $c(k) = \text{clip}(0.95 - 0.13k + \mathcal{N}(0, 0.06))$, falling below the standard 0.5 decision threshold around $k \geq 3$.

(S2) *Noise in both classes.* Hedged or hypothetical claims (considered and rejected by the text, hence not asserted) are flagged by NLI as contradicting asserted claims with confidence $w \sim U(0.55, 0.85)$; and spurious sub-threshold asserted–asserted contradiction edges occur with $w \sim U(0.10, 0.45)$.

Setup. Synthetic reasoning chains of six asserted claims connected by entailment edges ($w \sim \text{Beta}(20, 2)$); violated cases append an asserted statement contradicting claim A_0 at step distance $k \in \{1, \ldots, 5\}$, with the direct edge confidence drawn per (S1) and an optional weak paraphrase edge to A_1; both classes receive (S2) noise. 120 cases per (k, class) cell. Three detectors: Count (# contradiction edges with $w \geq 0.5$; the standard baseline), Sheaf (the unified energy of §3.3, assertion-aware, no thresholding), Sat (DPLL on edges with $w \geq 0.5$). Localization credit: for Count, the injected edge itself is above threshold; for Sheaf, the top-1 weighted-residual hotspot is the injected edge.

k	AUROC Count	AUROC Sheaf	AUROC Sat	Loc. Count	Loc. Sheaf
1	0.881	**1.000**	1.000	100.0%	**100.0%**
2	0.875	**1.000**	1.000	100.0%	**100.0%**
3	0.831	**1.000**	0.933	86.7%	**100.0%**
4	0.537	**0.993**	0.554	10.8%	**97.5%**
5	0.504	**0.970**	0.500	0.0%	**83.3%**

Table 1: Mechanism study: detection AUROC and top-1 localization rate versus violation distance k ($n = 120$ per cell). Threshold-based methods collapse to chance as the direct-edge confidence crosses the threshold; the weighted sheaf energy degrades gracefully and retains localization.

Findings. Two mechanisms produce the separation in Table 1. *Threshold collapse versus graceful degradation:* once (S1) pushes the direct-edge confidence below 0.5 ($k \geq 4$), Count and Sat lose the edge entirely and fall to chance, while Sheaf continues to accumulate the sub-threshold soft evidence as weighted energy (AUROC 0.970 at $k = 5$) and still localizes the injected edge in 83%

of cases where Count localizes 0%. *Hypothetical reconciliation:* even at $k = 1$–2, Sheaf leads by about 12 AUROC points because its global minimization sets non-asserted (hedged) claims to values consistent with the asserted graph, discounting the (S2) false-positive edges that Count tallies indiscriminately.

Scope of the claim. This is a simulation, not evidence about deployed systems. It establishes the conditional statement: *if* real NLI confidences decay with distance (S1) and produce hedged-claim false positives (S2)—both consistent with known NLI behavior, and both directly testable—*then* the sheaf energy separates structurally from counting. The pre-registered protocol of §7 tests exactly (S1)/(S2) and the induced separation with real models; the harness used here becomes the real experiment by replacing simulated confidences with model outputs. Success criteria are pre-registered in §7. A reference implementation of f_3 (weighted sheaf energy, convex minimization over the constraint polytope, DPLL certifier with deletion-minimal unsatisfiable cores, hotspot localization) and the study harness accompany the paper.

7 Pre-Registered Empirical Protocol

(1) Datasets and gate mapping. Format: structured-output benchmarks plus synthetic grammar corruption at controlled edit distances (ground-truth f_1). Fact: TruthfulQA and FEVER-style verification with a frozen retrieval index. Logic: GSM8K/MATH step-consistency with *adversarial contradiction insertion at known graph locations and controlled distance k*—the field version of §6. Alignment: human-adjudicated safety benchmarks. Intent: IFEval-style verifiable-constraint suites.

(2) Primary metrics. Per-gate AUROC and FPR at 95% TPR; empirical conformal coverage versus nominal α with Clopper–Pearson intervals (a direct field test of Theorem 4.2); ACI long-run error tracking under injected drift (Theorem 4.4); ECE of internal classifiers; end-to-end Dec accuracy against human triage.

(3) Ablations with pre-registered predictions. (a) MMD vs. Sinkhorn vs. plug-in W_2: plug-in W_2 degrades with embedding dimension per its $n^{-1/d}$ rate; the others do not. (b) *Sheaf vs. counting vs. SAT-only on real chains*—the empirical linchpin; pre-registered success criteria: AUROC gap ≥ 3 points overall or localization advantage ≥ 10 points, with the gap widening in k as in Table 1. (c) Lexicographic Π vs. full $K(y)$ for repair routing. (d) Mean-MMD vs. max-form f_5 on a single-negation adversarial suite: the mean form's miss rate tends to 1 as m grows, by Prop. 3.4.

(4) Assumption tests. Direct measurement of (S1): NLI confidence on true contradiction pairs as a function of reasoning distance; of (S2): NLI false-positive rate on hedged/hypothetical constructions.

(5) Dependence diagnostics. Empirical correlation matrix of $(f_1, \ldots, f_5)$; strong dependence motivates the safety-first fixed-sequence procedure of Theorem 4.3.

(6) Instrument audit. Human-annotated decomposition benchmark for ε_D (claim-level recall, Clopper–Pearson interval); edge-detection benchmark with adversarial paraphrase/negation for $\varepsilon_{\mathrm{NLI}}$; abstention floor $c_{\min}$ reported with the resulting abstention rate; all feeding the power certificate of Prop. 4.6.

(7) Drift monitoring. E-process monitor state and recalibration events logged; coverage audited *within* drift episodes, where the Regime-III guarantee is exactly the testable object.

8 Limitations and Open Problems

7.1 Instrument dependence. All distributional gates are relative to the fixed E, D; guarantees are for the scores as defined (Type-I end-to-end by Prop. 4.1; Type-II bounded by Prop. 4.6 and audited). Encoder-class invariance is open. **7.2 Assumption ledger.** Regime I requires exchangeability; Regime II requires estimable shift; Regime III guarantees long-run frequency only. Per-instance guarantees under fully adversarial shift are impossible in general; the fail-safe posture (Prop. 4.5) is the stated conduct in that gap. Marginal (not conditional) coverage is delivered; conditional coverage by prompt-difficulty strata via Mondrian conformal is the designated next theoretical step. **7.3 Relaxation gap.** A continuous zero of f_3 certifies only relaxed consistency (Prop. 3.2); the certifier closes the gap at worst-case exponential cost, and Prop. 3.3 shows this is unavoidable unless P = NP. **7.4 Mechanism-study scope.** §6 validates a conditional mechanism under (S1)/(S2), not a field claim; the protocol of §7 carries the burden of proof. **7.5 Lexicographic repair order.** A convention; no optimality claim is made.

Open Problem (Tarski unification). Recast the hard and soft logic layers as sheaves valued in complete lattices with the Tarski Laplacian [8], where global sections are fixed points and Boolean semantics is native.

9 Conclusion

Five well-posed, individually guaranteed instruments beat one undefined manifold. The framework presented here gives the guardrail architecture already running in production its missing mathematical contract: computable severities, finite-sample false-rejection control graded across three drift regimes, a complete and mutually exclusive error taxonomy, an unmaskable safety veto, and—in the sheaf logic gate—a candidate for genuinely new detection capability whose decisive test is pre-registered.

A Correction Table: v1 $\to$ v3

References

[1] A. Aho and T. Peterson. A minimum-distance error-correcting parser for context-free languages. *SIAM J. Comput.*, 1972.

[2] A. Andoni, A. Naor, and O. Neiman. Snowflake universality of Wasserstein spaces. *Ann. Sci. Éc. Norm. Supér.*, 2018.

[3] J. Cherian, I. Gibbs, and E. Candès. Large language model validity via enhanced conformal prediction methods. *NeurIPS*, 2024.

[4] CoFact: Conformal factuality guarantees for language models under covariate shift. *ICLR*, 2026.

[5] S. Coles. *An Introduction to Statistical Modeling of Extreme Values.* Springer, 2001.

[6] N. Fournier and A. Guillin. On the rate of convergence in Wasserstein distance of the empirical measure. *Probab. Theory Relat. Fields*, 2015.

[7] A. Genevay, L. Chizat, F. Bach, M. Cuturi, and G. Peyré. Sample complexity of Sinkhorn divergences. *AISTATS*, 2019.

v1 construct	**Defect**	**v3 replacement**
Riemannian "semantic space" with Fisher metric	Fisher metric requires a parametric family; the space was never constructed	Encoder embeddings; empirical measures (§2)
Fiber bundle with Karcher trivialization	Fiber/sheaf unspecified; uniqueness conditions absent	Removed; the surviving sheaf is the finite cellular sheaf of §3.3
W_2 projections onto convex sets (f_1, f_4)	Not well-posed in positively curved Wasserstein space	Exact edit distance (§3.1); calibrated classifier on raw text (§3.4)
Čech obstruction with Hodge norm (f_3)	Sheaf unspecified; infinite-dimensional Hodge theory undefined	Weighted cellular sheaf Laplacian with relax-and-certify (§3.3)
Isometric RKHS embedding of W_2 (f_5)	Impossible for $d \geq 3$ [2]	Kernel mean embedding, isometric w.r.t. MMD; max-form aggregation (§3.5)
Boundary-regularity conjecture via a reference measure on $\mathcal{P}(M)$	No canonical reference measure exists; Hausdorff bound meaningless in infinite dimensions	Atomless-score almost-sure well-posedness (Prop. 5.6)
Global Lipschitz equivalence of three norms	False; only one-sided bounds hold	Prop. 5.9, one-sided, interpretability-only
EVT thresholds with Fréchet law	Bounded scores lie in the Weibull domain; conditions unverified	Conformal theorems 4.2–4.4 across three regimes; EVT optional with diagnostics
Lexicographic Π as sole classifier	Masks safety violations behind lower-index failures	Parallel evaluation, safety veto, Π demoted to repair routing (§5.2)

[8] R. Ghrist and H. Riess. Cellular sheaves of lattices and the Tarski Laplacian. *Homology Homotopy Appl.*, 2022.

[9] I. Gibbs and E. Candès. Adaptive conformal inference under distribution shift. *NeurIPS*, 2021.

[10] M. Goemans and D. Williamson. Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming. *J. ACM*, 1995.

[11] A. Gretton, K. Borgwardt, M. Rasch, B. Schölkopf, and A. Smola. A kernel two-sample test. *J. Mach. Learn. Res.*, 2012.

[12] J. Hansen and R. Ghrist. Toward a spectral theory of cellular sheaves. *J. Appl. Comput. Topol.*, 2019.

[13] P. Liang et al. Holistic evaluation of language models. *TMLR*, 2023.

[14] S. Huntsman, M. Robinson, and L. Huntsman. Prospects for inconsistency detection using large language models and sheaves. arXiv:2401.16713, 2024.

[15] J. Zhou et al. Instruction-following evaluation for large language models. arXiv:2311.07911, 2023.

[16] C. Mohri and T. Hashimoto. Language models with conformal factuality guarantees. *ICML*, 2024.

[17] T. Rebedea et al. NeMo Guardrails: A toolkit for controllable and safe LLM applications. *EMNLP Demo*, 2023.

[18] R. Tibshirani, R. Foygel Barber, E. Candès, and A. Ramdas. Conformal prediction under covariate shift. *NeurIPS*, 2019.

[19] V. Vovk. Testing randomness online. *Statist. Sci.*, 2021.

[20] Y. Abbasi Yadkori et al. Mitigating LLM hallucinations via conformal abstention. arXiv:2405.01563, 2024.